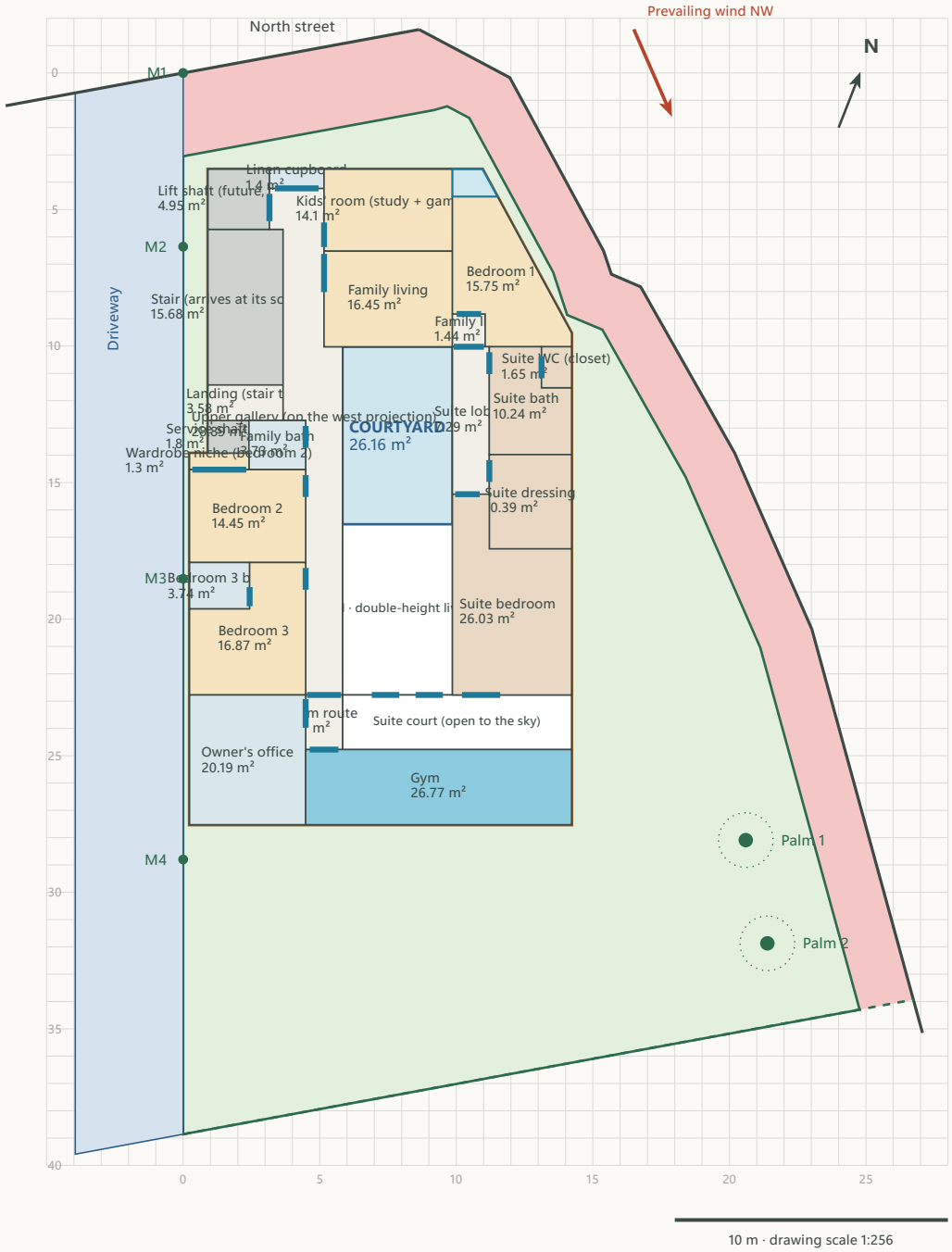


10 m · drawing scale 1:256

Taif / Palm House

First floor V5.1 · split projection (0.675 m in + 0.675 m out)

24 Oct 2026 · on ground floor V5.1 · rooms in data/rooms_ff.csv · design and furniture in studies/ff_family.typ and studies/ff_suite.typ · schedule and light check on the next pages

First floor V5.1 (split projection). The V5.0 1.35 m projection is split: **0.675 m inward** over the courtyard and the double-height void (x 4.25–4.925, a plain cantilever, **no columns**) and **0.675 m outward** beyond the GF west wall (y 10.4–24, the west wall on cantilever beams); the whole west band moves 0.675 m west. FF west wall face **0.11 m from the M line** (GF 0.785 m); courtyard sky and void **4.025 m** on axes (V5.0 3.35). The gallery (1.35 m) and the west rooms (4.25 m) keep their width: **bedroom 2**, **bedroom 3 with its own bath**; the **family bath** narrows to 2.075 m (the shaft is fixed). North band: **kids' room (study + gaming)** and the **family living** on the courtyard edge. **Bedroom 1** is one room on the straight y 6.5 line. The **L**: **owner's office** (west), **gym route**, **gym**, the **suite court** open to the sky in front of the clerestory windows, and the **master suite** (lobby, luxury bath with WC closet, walk-in dressing, bedroom). The east duct is removed. Stair, lift, landing and west shaft stay; the west face steps out at y 10.4. Colours: suite (beige) · family (yellow) · baths, office and linen (blue-grey) · gym (blue) · circulation (light) · fixed (grey) · courtyard, void and court (white / blue). Blue marks: doors and the two clerestory windows. Areas: available 243.75 m² · rooms 226.97 m² (circulation 43.9 m²) · suite court 16.77 m².
✓ **data/rooms_ff.csv matches the two studies room for room.**

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**318.86 m²**
Other open voids	**41.93 m²**
Courtyard (open to sky)	**26.16 m²**
Rooms placed (23)	**249.4 m²**
Limit minus courtyard and voids	**250.77 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**1.37 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Stair (arrives at its south end)	M	2.75 × 5.7	15.68 m²
Lift shaft (future, 2 stops)	M	2.25 × 2.2	4.95 m²
Service shaft (west)	M	1.5 × 1.2	1.8 m²
Landing (stair top)	M	2.75 × 1.3	3.58 m²
Upper gallery (on the west projection)	M	2.675 × 18.55	28.89 m²
Linen cupboard	M	2 × 0.7	1.4 m²
Family bath	M	2.075 × 1.8	3.73 m²
Wardrobe niche (bedroom 2)	M	2.175 × 0.6	1.3 m²
Bedroom 2	M	4.25 × 3.4	14.45 m²
Bedroom 3 bath	M	2.2 × 1.7	3.74 m²
Bedroom 3	M	4.25 × 4.85	16.87 m²
Kids' room (study + gaming)	M	4.7 × 3	14.1 m²
Family living	M	4.7 × 3.5	16.45 m²
Family link	M	1.2 × 1.2	1.44 m²
Bedroom 1	M	4.361 × 5.5	15.75 m²
Gym route	M	1.35 × 2	2.7 m²
Owner's office	M	4.25 × 4.75	20.19 m²
Gym	M	9.736 × 2.75	26.77 m²
Suite lobby	M	1.35 × 5.4	7.29 m²
Suite bath	M	3.011 × 3.95	10.24 m²
Suite WC (closet)	M	1.1 × 1.5	1.65 m²
Suite dressing	M	3.011 × 3.45	10.39 m²
Suite bedroom	M	4.361 × 7.35	26.03 m²

First floor V5.1 · room schedule

wall centre lines (axes) · clear ≈ axes minus half a wall on every edge (from the studies) · CSV coordinates

Room	Part	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	family	2.75 × 1.3	3.57	3.14
Upper gallery (on the west projection)	family	polygon	28.89	25.92
Linen cupboard	family	2 × 0.7	1.4	–
Family bath	family	2.08 × 1.8	3.74	2.96
Wardrobe niche (bedroom 2)	family	2.17 × 0.6	1.3	–
Bedroom 2	family	4.25 × 3.4	14.45	12.84
Bedroom 3 bath	family	2.2 × 1.7	3.74	2.92
Bedroom 3	family	polygon	16.87	14.97
Kids' room (study + gaming)	family	4.7 × 3	14.1	12.44
Family living	family	4.7 × 3.5	16.45	15.16
Family link	family	1.2 × 1.2	1.44	1.08
Bedroom 1	family	polygon	15.75	13.83
Gym route	L	1.35 × 2	2.7	2.03
Owner's office	L	4.25 × 4.75	20.19	18.16
Gym	L	9.74 × 2.75	26.77	23.96
Suite court (open to the sky)	L	8.39 × 2	16.77	–
Suite lobby	L	1.35 × 5.4	7.29	5.94
Suite bath	L	polygon	10.24	8.79
Suite WC (closet)	L	1.1 × 1.5	1.65	1.09
Suite dressing	L	3.01 × 3.45	10.39	9.01
Suite bedroom	L	polygon	26.03	23.56
Rooms (indoor)			**226.97**	
Suite court (open)			**16.77**	
Stair, lift, west shaft, air tower			23.8	

Key numbers

- **Master suite:** 55.6 m² indoors (lobby, bath + WC closet, dressing, bedroom), + the suite court 16.77 m².
- **Circulation:** 43.9 m² (landing, gallery incl. the projection, family link, gym route, suite lobby).
- **Courtyard sky** 4.025 × 6.5 m and **double-height void** 4.025 × 6.25 m on axes (V5.0 3.35, GF 4.70); the ground-floor courtyard and living keep their full floor.
- **Bedrooms 2 and 3:** 4.25 m wide on axes (≈ 4.0 m clear), 0.675 m further west. **Family bath** 2.075 m wide (V5.0 2.75): the shaft is fixed.
- **FF west wall** 0.11 m from the M line; on cantilever beams y 10.4–24.

Where the detail is

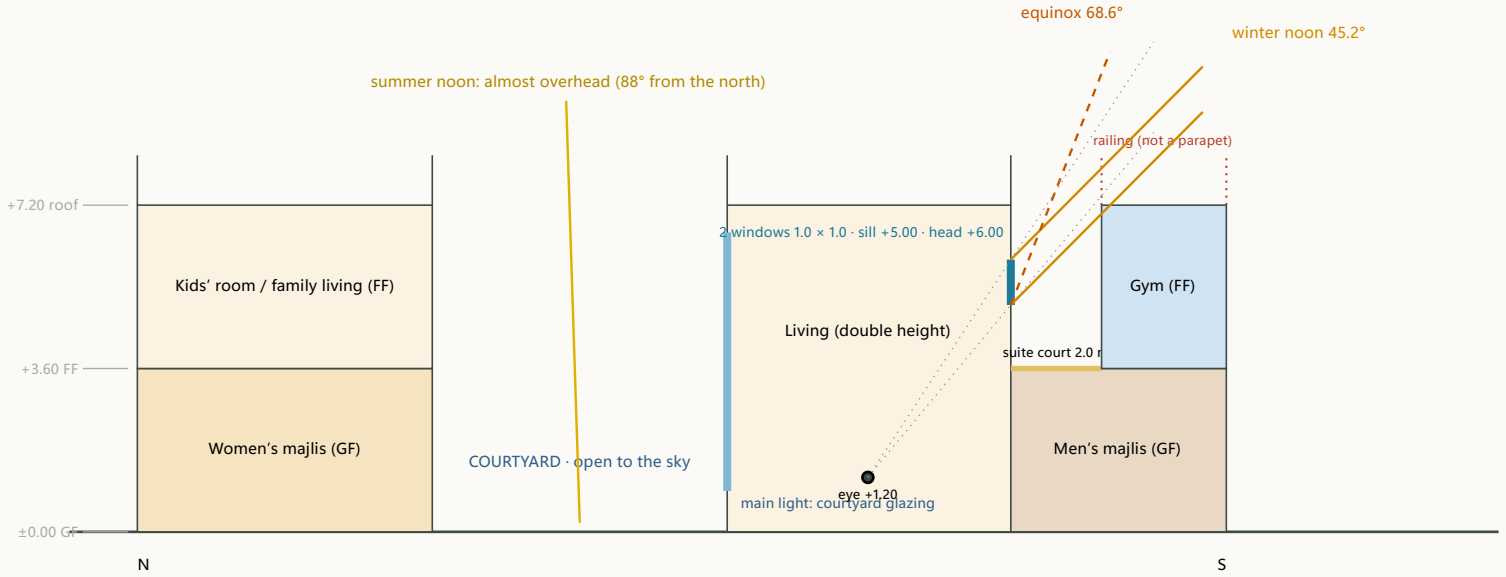
- **Furniture, doors, windows, drains, walking distances, routes ≥ 1.0 m and room checks:** studies/ff_family.typ (part 1) and studies/ff_suite.typ (part 2).
- **Construction notes** (projection, columns, east duct, sunken baths, suite court, cupboards, loads): docs/DETAILS.md.
- **Decisions and open items:** docs/FIRST_FLOOR_BRIEF.md (V5.1) and docs/notes.md (Changes in V5.1).

Open items

- **TOP OPEN ITEM, before construction drawings:** municipality / engineering office to confirm that an upper-floor projection 0.11 m from the M line is allowed. If not, go back to the V5.0 solution (full 1.35 m inward projection on three columns at x 5.6, tag v5.0).
- Engineer: the FF west wall on cantilever beams (0.675 m, y 10.4–24: beam depth, wall load, deflection); the inward 0.675 m cantilever with no columns; loads and vibration of the gym and walls over the men's majlis, the y 3.0 partition over the women's majlis.
- MEP: the stack outside the east wall (suite bath, vent stack), the kitchen extract through the east wall, the hood position south of y ≈ 7.5.
- The clerestory sill vs the suite court floor (eye height, or obscured glass); a daylight check of the narrower courtyard.

First floor V5.1 · light check · the double-height living's south windows

section north → south through the middle band (CSV x ≈ 7.3) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.10 · a full daylight simulation comes later



Behind the 2.0 m suite court	Winter noon sun on the windows	Equinox noon sun	Sky seen through the windows from the eye
gym with a 1.10 m parapet (top +8.30)	0 %	100 %	42 %
gym with a railing only (solid top +7.20) · **V5.1**	81 %	100 %	100 %

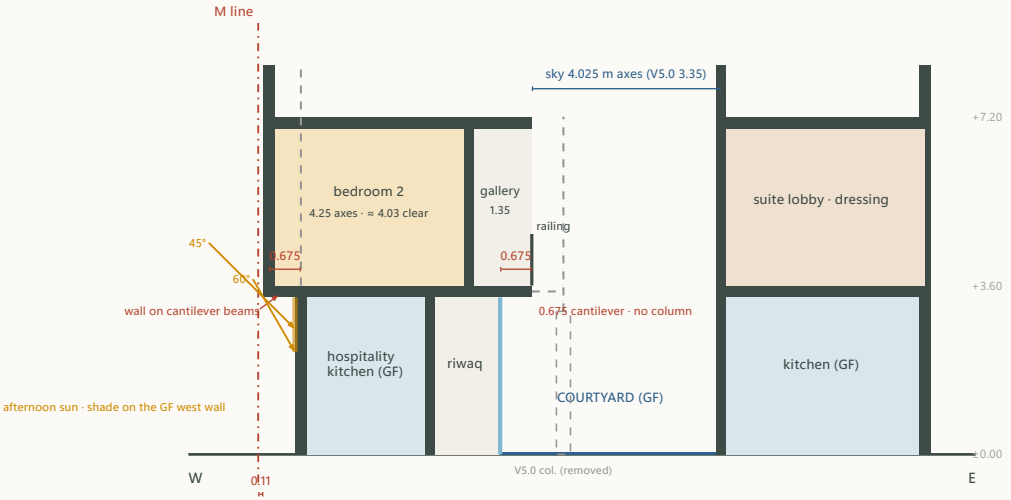
How the percentages are defined. "Sun on the windows": the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox), in this section; the ray from each point of the window must pass over the top of the gym's north wall 2.0 m away. "Sky seen": from a seated eye 1.20 m high in the middle of the double-height living floor (CSV y 16.1), the share of the vertical angle of the windows that shows sky rather than the gym. Pure geometry: no reflections, no sky model.

- **The main light of the double-height living comes from the courtyard.** The two small south windows (1.0 × 1.0 m, sill +5.00) add light and let warm air out high on the south.
- **V5.0:** the 2.0 m strip in front of them is now the **suite court**, walled on four sides and open to the sky; the gym's north wall is its south side, ending in a **railing, not a parapet**.
- **Privacy:** the windows open onto the suite court, so their sill stays above eye level from the court floor (or obscured glass); to check against the floor levels.
- **Summer:** the sun is almost overhead; light-coloured court walls and small hoods (≈ 0.4 m) over the windows.

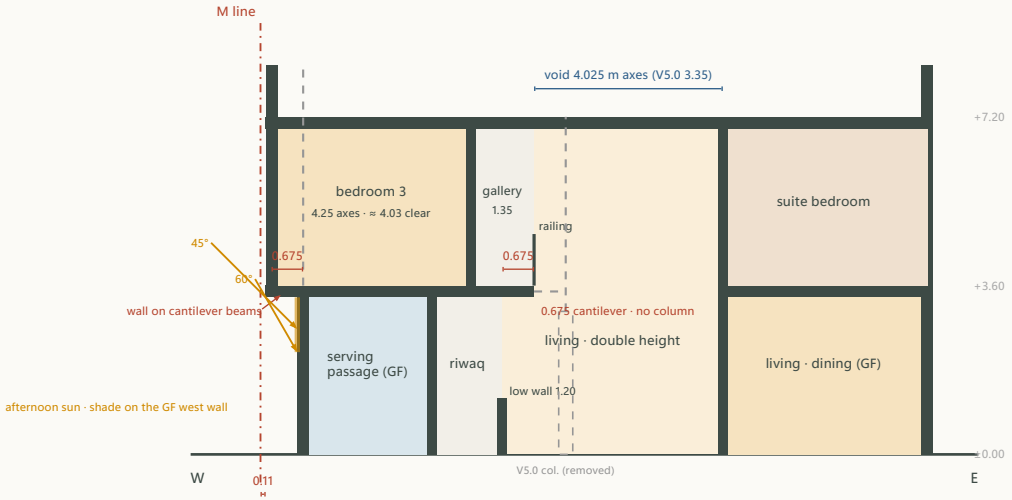
V5.1 · split projection · against V5.0 (record of the decision)

sections looking north, CSV x · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · grey dashed = V5.0 (FF west wall on x 0, gallery edge beam and column on x 5.6)

A–A · through the courtyard (y 12.0)



B–B · through the double-height living (y 16.5)



Item	V5.0	V5.1 · split	Change
Courtyard sky opening, axes	3.35 m	4.025 m	+0.675
Courtyard sky opening, ≈ clear	3.15 m	3.825 m	+0.675
Double-height void width, axes (≈ clear)	3.35 (3.15) m	4.025 (3.825) m	+0.675
West room width, axes (≈ clear)	4.25 (4.025) m	4.25 (4.025) m	=
Gallery width, axes (≈ clear)	1.35 (1.225) m	1.35 (1.225) m	=
Columns in the courtyard / living	yes · 3 on x 5.6 + edge beam	**no** · plain 0.675 m cantilever	−3
FF west face → M line	0.785 m (on the GF wall)	**0.11 m**	−0.675 m
Walls on cantilever	none (gallery railing on the edge beam)	FF west wall y 10.4–24 + returns: 14.95 m	new
FF outline (axes)	309.68 m²	318.86 m²	+9.184 m²
Courtyard sky + double-height void	42.71 m²	51.32 m²	+8.609 m²
FF floor (outline − voids)	266.97 m²	267.55 m²	+0.575 m²
FF rooms indoor (incl. circulation)	227.75 m²	226.97 m²	−0.775 m²
· circulation	45.72 m²	43.9 m²	−1.825 m²
· family bath	4.95 m² (2.75 wide)	3.74 m² (2.075 wide)	−1.215 m²
· gym	24.92 m²	26.77 m²	+1.854 m²
· wardrobe niche (bedroom 2)	0.9 m²	1.3 m²	+0.405 m²
Suite court (open to the sky)	15.42 m²	16.77 m²	+1.352 m²

V5.0 from git tag v5.0; variant from data/rooms_ff.csv (live). Clear ≈ axes − ½ wall each side (250 outer, 200 inner, ≈ 50 mm railing on the void edge); sky opening clear = axes − 0.10 east wall − 0.10 slab edge with railing.

- **Split:** band 4.25 + 1.35 = 5.6 m kept. Rooms ≥ 4.0 m clear needs ≥ 4.225 m on axes, gallery ≥ 1.0 m clear ≥ 1.125 m: both hold for any split. The site caps the outward part at 0.785 m (GF face to the M line); the **even split 0.675 / 0.675** leaves 0.11 m.
- **Structure:** the inward 0.675 m cantilever (back span 1.5 m to x 2.75) carries only the gallery and its railing: **no columns needed** (to confirm); the three V5.0 columns at x 5.6 are removed from both floors. The FF west wall (y 10.4–24) stands on **cantilever beams** 0.675 m beyond the GF west wall: engineer to size beam depth, wall load and deflection.
- **Climate:** the outward projection shades the top of the GF west wall from the afternoon sun (≈ 0.675 × tan of the sun's profile angle: 0.7 m at 45°, 1.2 m at 60°) and roofs the guest entrance.
- **Top open item:** FF west face 0.11 m from the M line: municipality / engineering office to confirm an upper-floor projection this close is allowed; if not, back to V5.0 (tag v5.0).
- **Forced changes:** family bath 2.75 → 2.075 m wide (shaft fixed); wardrobe niche 1.5 → 2.175 m; gallery 2.18 m at the stair top; gym +0.675 m, suite court +0.675 m; west doors, windows and furniture 0.675 m west. Everything else as V5.0.